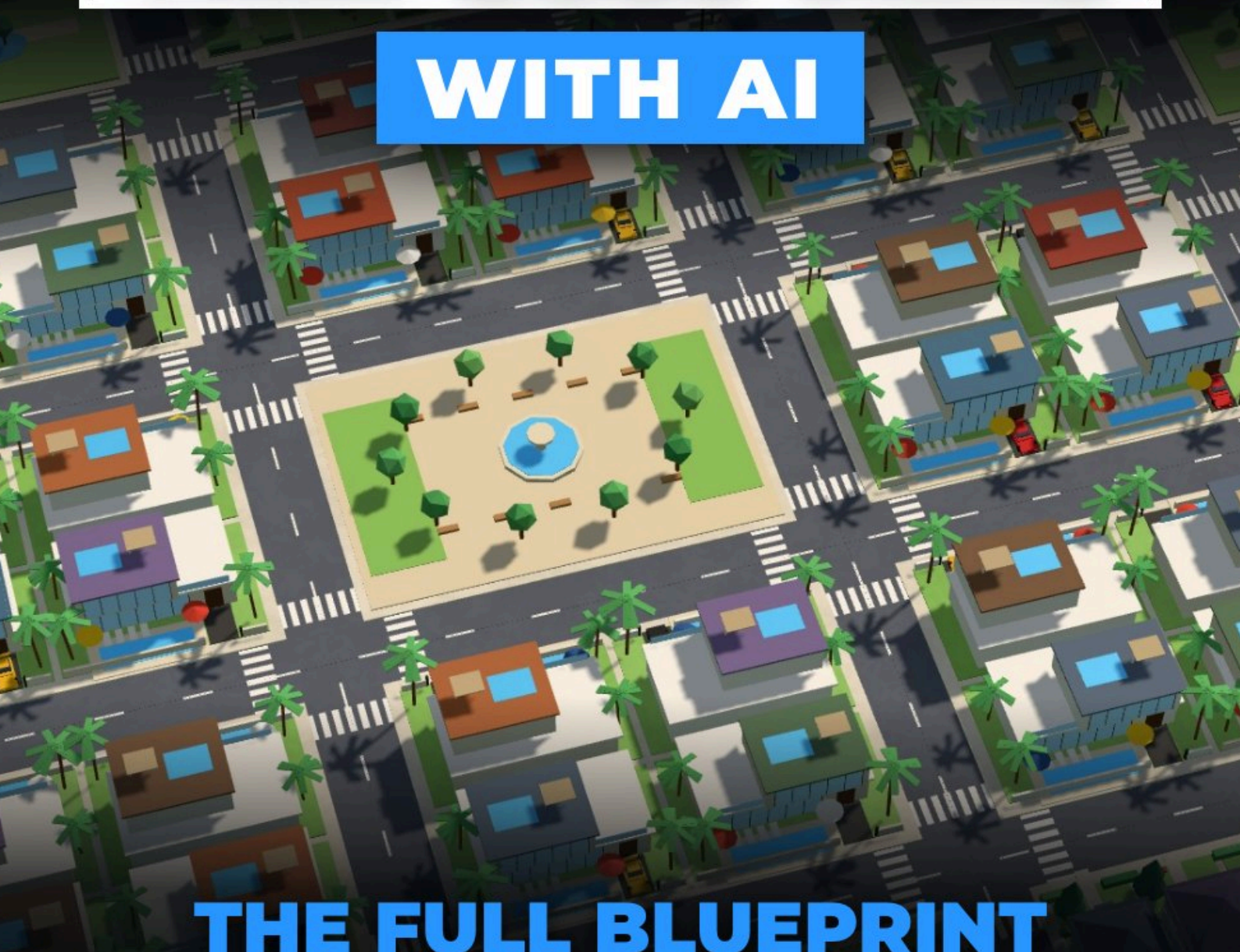
BUILT WITH CLAUDE, ONE PROMPT AT A TIME

HOW TO BUILD A 3D CITY THAT GOES VIRAL WITH AI



THE FULL BLUEPRINT

EVERY PROMPT AND EVERY FILE,
FROM THE CITY TO THE DMS

OMNYA LAB

START HERE

WHAT YOU'LL HAVE AT THE END

- ✓ A cozy 3D city in the browser, 1,359 lots, villas downtown and trailers at the edge, written by Claude in three.js.
- ✓ A house for every person, drawn from their username: the same name always gives the same house.
- ✓ A 37 second vertical reel, rendered frame by frame from that city, with voice and sound effects.
- ✓ The cover and the caption that ask for one word in the comments.
- ✓ A live map on your site: comment, DM, one-time link, the house appears, then the offers.

WHAT YOU NEED

Claude Code
It reads this PDF, writes the city, the reel page and the site pages with you.

ManyChat Pro
For the comment, the follow check and the DM with the link.

ElevenLabs
The voice. We used Archie.

Vercel + Upstash
The site and a small Redis, both on the free plans.

CHAPTER 00

THE FAST WAY
CLAUDE CODE

Open Claude Code in an empty folder, drop this PDF in it and paste the prompt below. Claude reads the PDF page by page, asks you about your version and builds it with you one step at a time.

```
I bought the Full Blueprint "How to build a 3D city that goes viral with AI" by Omnya Lab. The PDF is in this folder: read it page by page, from chapter 01 to chapter 09, before you do anything.

I want what Omnya Lab did with Omnya City (instagram.com/omnyalab): a cozy 3D city in three.js, a house for every person who comments, drawn from their username, a 35 to 40 second reel rendered frame by frame from the browser, and a live map where people reserve their house from a one-time link in their DMs.

First ask me about my version: whose city it is (my followers, my buyers, my guests), the look, the reward, the comment word. Then follow the PDF in order. After every step show me the result in the browser and wait for my OK before the next one. Use the prompts in the PDF as they are, changing only the parts about my brand.
```

This prompt is the one we wrote for you. Every other prompt in this PDF is a request we really sent while building Omnya City, translated from Italian, with nothing added.

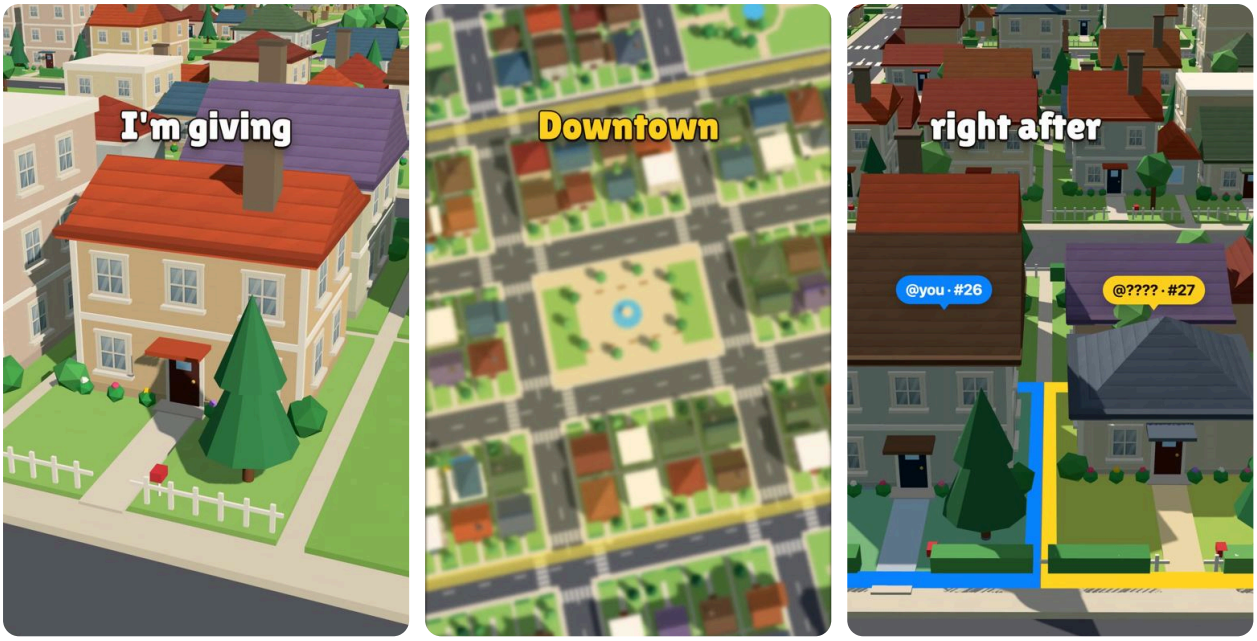
CHAPTER 01

THE FORMAT
AND THE STORY

Omnya City started as a format we saw working on another account: a city where every follower gets a house. Nico's first message was to copy the format, not the creator: "we have to make them the same but for Omnya". New name, new city, new look, our own story.

Write the story first, as the voice will say it, then build only what the story needs. Ours is eight beats in 37 seconds:

BEAT	THE VOICE	THE PICTURE
Hook	I'm giving every follower a free house in my city.	One cozy house, close, the camera already moving
The rule	But where you live depends on when you follow.	The camera rises over the whole town
Early	Early? Downtown.	A yellow frame around the centre, then the fountain
Late	Late? …suburbs.	Trailers and shacks at the edge
Username	Your username designs your house. No two are the same.	A search bar types a name, the house changes
Neighbour	And your neighbour? Whoever comes right after you.	Two lots outlined, @you #26 and @???? #27
Growth	Every milestone, the city grows.	Towers rise in the centre
Call	Comment "house", I'll DM you yours. House #1 is still empty. Downtown won't last.	The comment, the DM, then the empty lot with the pin



THREE FRAMES OF THE REEL: THE HOOK, EARLY, THE NEIGHBOUR

> Why it works: it is a free gift with a rule people want to beat (“early is downtown”), a reason to comment now (house #1 is still empty) and a reason to tag a friend (the neighbour). Every comment is a follower, and every follower is a name on your map.

CHAPTER 02

THE CITY IN CODE

THREE.JS, ONE FILE

The whole city is one JavaScript file, `city.js`, with one function, `buildCity(scene, style)`. It returns the list of lots and a `makeHouse()` to draw any house anywhere. This is the first prompt, the requests that shaped it, put together:

PROMPT 1 · CLAUDE CODE · THE CITY

```
Build a cozy 3D city in three.js that runs in the browser, seen from above like a
cozy game.

It must not be round and not a perfect grid: real streets, with roads that lead
out of town, and blocks that are not all the same, some long and horizontal, some
more vertical.

Whoever arrives first gets a villa in the centre, the next ones smaller villas,
then normal houses, and at the edge, in the suburbs, trailers and shacks. Every
house different.

Colours very real but cozy. Textures on the roofs, chimneys, windows like a cozy
game, soft shadows that all come from the same sun. Green hills around the city, a
nice blue sky, the sea on one side.

When the camera rises the roofs must stay clean: no flicker, no blue and white
shimmer.
```

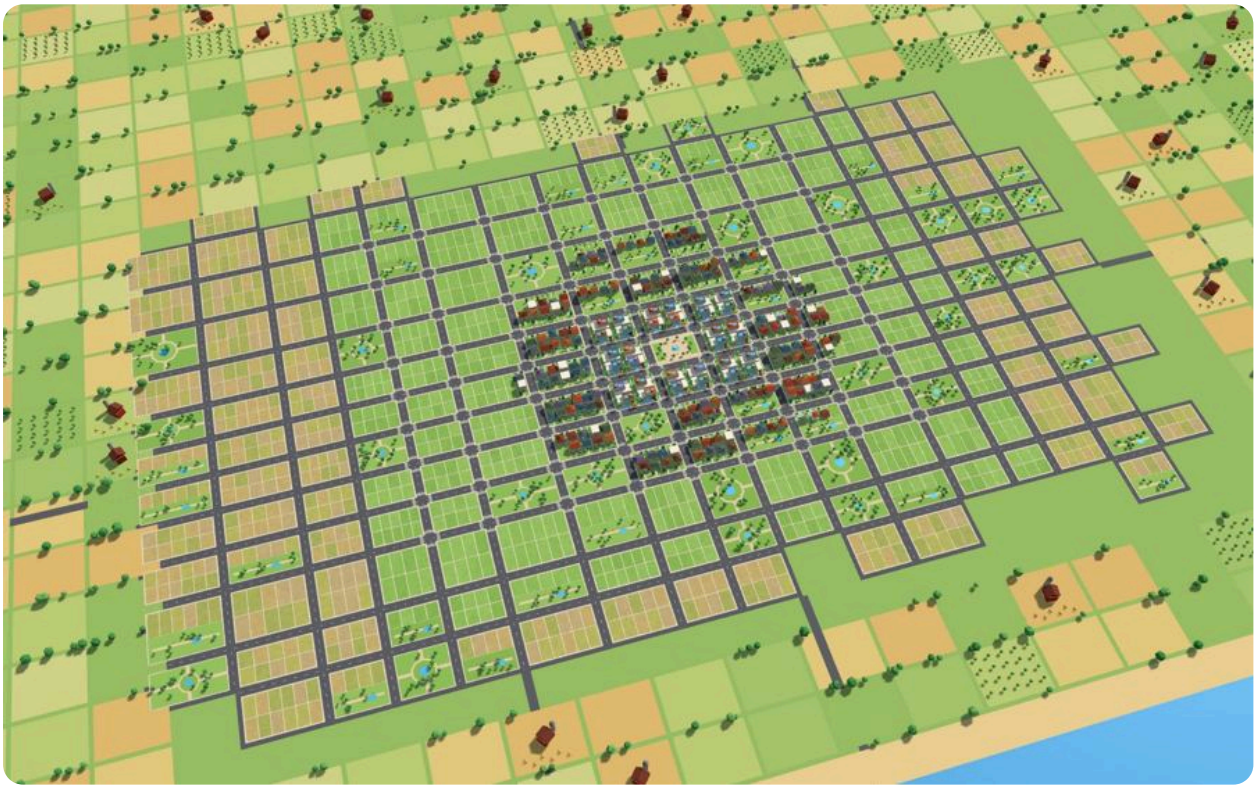
WHAT CAME OUT

- A grid of 17 × 13 blocks with the corners cut and the edge irregular, so the town is never a square or a circle.
- Blocks of different widths: the three columns round the centre are fixed, the others take 34, 44, 56 or 62 metres at random.
- Three to six houses per row depending on the width, sometimes half a block left as a pocket park, about one block in nine a park with a pond.
- Outside the town: fields, red barns with silos, orchards, green hills with woods, the sea and the beach on one side.
- Every random choice comes from one seeded generator, so the city is the same on every phone, every time.

THE RINGS DECIDE THE HOUSES

The distance from the fountain, in blocks, picks the type. Twelve types, weighted per ring: villas only in rings 0 and 1, trailers and shacks only from ring 4.

```
const MIX = [   // ring 0 is the fountain, ring 7 the edge of town
  { villaBig: 10, villaSmall: 8, gable2: 22, hip2: 18, flat2: 14, L2: 16,
gable1: 6 },
  { villaBig: 10, villaSmall: 8, gable2: 22, hip2: 18, flat2: 14, L2: 16,
gable1: 6 },
  { villaSmall: 8, gable2: 22, hip2: 16, flat2: 12, L2: 18, gable1: 14, hip1:
10 },
  { villaSmall: 4, gable2: 18, hip2: 12, flat2: 10, L2: 16, gable1: 24, hip1:
16 },
  { gable2: 12, gable1: 28, hip1: 22, L1: 18, flat1: 10, trailer: 5, shack: 5
},
  { gable2: 6, gable1: 26, hip1: 22, L1: 14, flat1: 8, shack: 12, trailer: 12
},
  { gable1: 22, hip1: 16, L1: 8, shack: 26, trailer: 28 },
  { gable1: 12, hip1: 10, L1: 5, flat1: 4, shack: 33, trailer: 36 },
];
```



THE WHOLE TOWN FROM ABOVE

CORRECTIONS THAT CHANGED IT

- *“It must not be round, it needs streets and turns that lead out, like a real small town.”*
- *“The lots not all the same: some horizontal, some more vertical, just not round or diamond-shaped.”*
- *“Villas only at the start, the first ones to arrive, then small villas, and so on.”*
- *“Put both trailers and shacks when you frame the suburbs.”*

CHAPTER 03

THE COZY LOOK

THE DIFFERENCE IS IN THE DETAILS

The first versions were clean and dead. What made them cozy was one method: put a frame of ours next to a frame of the reel we were matching, and ask for exactly the difference. Nico's notes, in order:

- *"Look at his compared to ours: textures, shadows. We have to make it as beautiful as his."*
- *"He has houses on different levels, colours very real but cozy. Look at his windows compared to ours."*
- *"He even has the chimney, nicer shadows, really a cozy game."*
- *"Don't put the mountains in the fog at the back: make them nice and green, and let the sky be a nice blue."*
- *"When you raise the camera the roofs flicker, blue and white. It has to be clean."*

WHAT EACH NOTE BECAME

NOTE	IN THE CODE
Textures	A small shader on the materials: roof tiles in rows, siding lines on the walls, noise on the grass, paving squares on the sidewalks, grain on the asphalt.
Windows	One canvas texture with four panes, a reflection stripe and a frame, used on every window.
Colours	Nine wall colours from sand to grey-blue, nine roof colours from terracotta to slate, three lawn greens.
Shadows	One sun only, low from one side, 4096 shadow map in the reel. Chimneys on about a third of the houses.
Flicker	A logarithmic depth buffer, the roof pattern fading out when it gets smaller than a pixel, and the small details hidden when the camera is high.



DOWNTOWN: THE VILLAS TAKE TWO LOTS, SOME WITH A SUPERCAR OUTSIDE

REQUEST · CLAUDE CODE · DOWNTOWN

```
Downtown the villas are huge and take two lots. Some of them, not all, have a car
outside: one that looks like a red Italian supercar, another like a yellow one.
Similar, not the real brands.
```

Keep it a look-alike: no real logos or names on the cars. The shape and the colour are enough.

CHAPTER 04

HOUSES FROM USERNAMES

SAME NAME, SAME HOUSE

PROMPT 2 · CLAUDE CODE · THE HOUSES

```
Every follower gets a free house in the city.

Their Instagram username designs it: the same name always gives the same house,
and no two are the same. Turn the username into a number and use it as the seed
for the type of house, the walls, the roof and the door.

Where it goes depends on when they arrive: the first ones downtown by the
fountain, the late ones in the suburbs. Your neighbour is whoever comes right
after you.
```

The username is turned into a number with a small hash, and that number seeds a random generator. The generator picks the type for the ring, the walls, the roof, the door, the chimney, the garden. Nothing is saved but the name: the house is drawn again, identical, every time.

```
export function hashStr(s) {
  let h1 = 0xdeadbeef ^ 0, h2 = 0x41c6ce57 ^ 0;
  for (let i = 0; i < s.length; i++) {
    const ch = s.charCodeAt(i);
    h1 = Math.imul(h1 ^ ch, 2654435761);
    h2 = Math.imul(h2 ^ ch, 1597334677);
  }
  h1 = Math.imul(h1 ^ (h1 >>> 16), 2246822507) ^ Math.imul(h2 ^ (h2 >>> 13),
3266489909);
  return h1 >>> 0;
}
export function rng(seed) {             // the same seed always gives the same
numbers
  let a = seed >>> 0;
  return () => {
    a |= 0; a = (a + 0x6D2B79F5) | 0;
    let t = Math.imul(a ^ (a >>> 15), 1 | a);
    t = (t + Math.imul(t ^ (t >>> 7), 61 | t)) ^ t;
    return ((t ^ (t >>> 14)) >>> 0) / 4294967296;
  };
}
// the house of @sofia.m, anywhere, on any phone:
const house = city.makeHouse(null, hashStr('@sofia.m'), dist, lotWidth);
```



@LUCA_92, @THE.ARCHI, @SOFIA.M

> On the site the database keeps three things per person: the username, the lot number, the date. That is why the map loads fast with thousands of houses: they are all built from the names, in one batch, in the phone.

CHAPTER 05

THE REEL

FRAME BY FRAME FROM THE BROWSER

PROMPT 3 · CLAUDE CODE · THE REEL

```
Make the reel as an HTML page and render it frame by frame, 1080 x 1920, 30 fps.

Every camera move starts on the word. No scene is ever completely still: always a
slight move or a slow zoom. Every climb, drop and transition is super smooth, one
soft curve upwards, never sliding left and right or going diagonal.

Voice: Archie on ElevenLabs, normal and clean, no music. One opening sound effect
right at the very start, the effects a little lower than the voice.
```

HOW IT IS MADE

1. The reel is one HTML page: the city, a camera that follows a timeline, the words that come in on the voice, the phone screens drawn in HTML.
2. The voice is recorded first (Archie on ElevenLabs) and cut into 18 segments with their times, so every move can start on its word.
3. A Python script opens the page in Chrome with Playwright, moves the timeline to each frame and takes a picture: about 1,140 frames for 38 seconds.
4. The audio is mixed in Python: the voice at 1.3, each sound effect at half its level, a limiter at 0.95. No music.
5. ffmpeg puts the frames and the mix together, 1080 × 1920, 30 fps.

```
await pg.goto(url)                                    # the reel page, 1080 x
1920
await pg.wait_for_function('window.ready === true')   # the city is built
n = int(END * 30) + 1                                 # 30 frames a second
for f in range(n):
    await pg.evaluate(f'seek({f/30})')                # every tween and camera
move at time t
    await pg.screenshot(path=f'frames/{f:05d}.png')
```

THE RULES THAT CAME FROM THE EDITS

- *“There must be no completely still scenes, always a slight move or a slight zoom.”*
- *“All the climbs, drops and transitions super smooth, not going right and left while they rise or diagonally: a nice chill curve upwards.”*
- *“The first sound effect right at the start, an opening one, then lower them all a bit and raise the voice.”*

- *“Maybe the first part with the house lasts too long.” The hook must move from frame one.*

It took 24 versions. Render a contact sheet of one frame per second after every version: it shows timing mistakes faster than watching.

CHAPTER 06

COVER AND CAPTION

ONE WORD TO COMMENT

The cover is the town from above with three lines: “I’m giving every follower a FREE house”. “every follower” in yellow, FREE as a red stamp, house in white and big. Nothing else.



The caption opens with the question people have, then answers it, then gives the word. This is the one we posted:

```
Don’t miss your house! 🏠👇

How do I reserve my home, and where does it
go?

Just comment "HOUSE" or "🏠" and I’ll send
you a DM with your link. Open it, tap
Reserve my spot, type your Instagram name
and your house appears on the blueprint of
the city, right on your lot.

Where it lands depends on when you reserve.
The first ones get downtown by the
fountain, the late ones end up in the
suburbs, so don’t wait too long. Your
username designs the house, no two are the
same, and your neighbour is whoever comes
right after you.

Make sure you follow first, or my DM can’t reach you.

House #1 is still empty. Downtown won’t last.
```

Two words in quotes (“HOUSE” and the emoji) so people can comment from a phone without typing. The last line is the urgency, and it is true.

CHAPTER 07

THE LIVE CITY

COMMENT, DM, HOUSE

The reel promises a house, so the house has to exist the moment someone asks. Instagram no longer lets normal users log in to other sites, so the name comes from ManyChat, and the link it sends works once. Nobody can fill your city with fake names.

THE MANYCHAT FLOW

1. Trigger: a comment on the next post with HOUSE or 🏠. Three public replies at random, short.
2. First DM with a button, “Get my house”. Instagram delivers the next messages only after a tap.
3. Condition: follows your account. If not, “follow us, then tap Done”, and Done checks again.
4. External Request, POST to your site, header with your secret, body with the Instagram username and the contact id. The answer’s `url` goes into a user field, `city_url`.
5. Last DM: “Here’s your link”, with the button Reserve my spot opening `{{city_url}}`.

Use the address of your site that does not redirect (for us, with www): an External Request does not follow a redirect.

PROMPT 4 · CLAUDE CODE · THE SITE

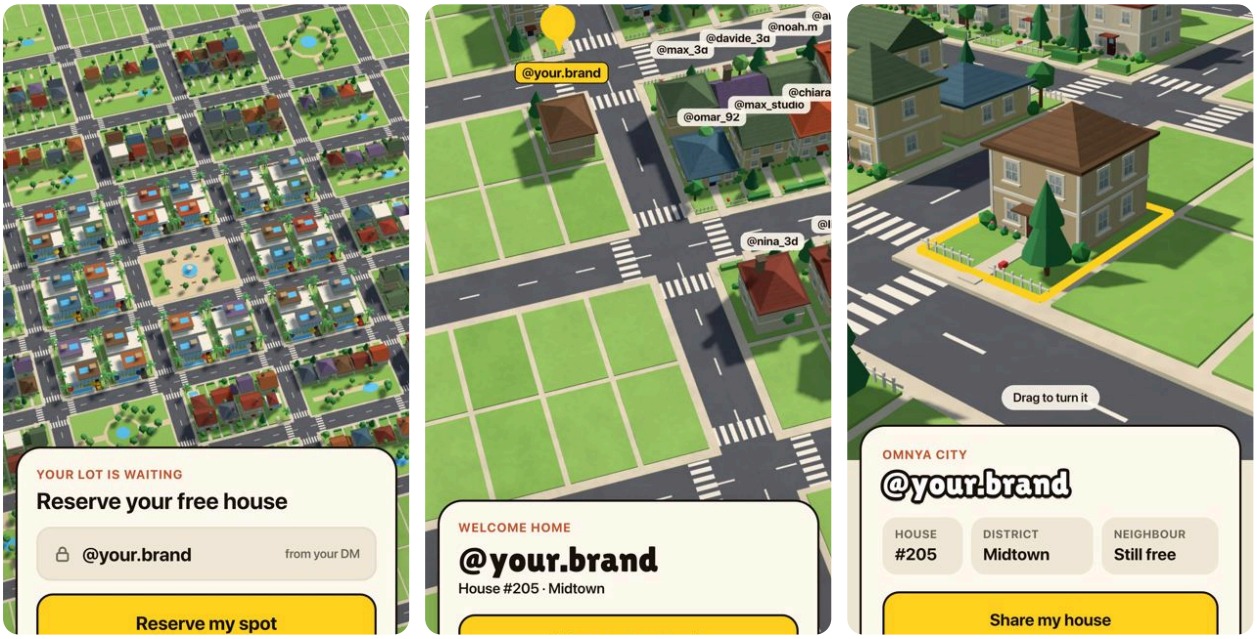
```
Make the page for the city on my site.

1. POST /api/city/link, protected by a secret in a header: it gets {username,
igId} from ManyChat, makes a one-time token in Redis tied to that profile and
answers {url: "https://mysite.com/city?k=TOKEN"} in real time. If that username
already has a house, it answers with the link to the house.
2. /city: the map from above like in the reel, booked houses drawn, free lots
empty, lots numbered from the centre outwards. With a valid ?k= it shows "Reserve
my spot" with the username already written and locked, no free text field: one
profile, one house. A field "I already have a house" to find a house by name.
3. POST /api/city/reserve {k}: gives the next free lot atomically, saves it, burns
the token, refuses doubles. GET /api/city/houses for the map.
4. /city/@username: the house in 3D that turns with a finger, its number, its
district, the neighbour, a share button, then the offers.
```

THE HEART OF IT

Reserving a lot must be safe when hundreds of people tap at the same second. Three Redis commands do it: GETDEL reads and burns the link in one step, SET NX gives one house per name, INCR hands out the next lot.

```
export async function reserve(token) {
  const t = await cmd('GETDEL', `city:tok:${token}`);        // read and burn
the link in one step
  if (!t) return { error: 'used' };
  const { u, ig } = JSON.parse(t);
  const lock = await cmd('SET', `city:u:${u}`, 'pending', 'NX');  // one
profile, one house
  if (lock !== 'OK') return { house: await houseOf(u), fresh: false };
  const n = Number(await cmd('INCR', 'city:n'));              // the next lot,
never twice
  const house = { u, n, t: Date.now(), ig };
  await cmd('SET', `city:u:${u}`, JSON.stringify(house));
  await cmd('ZADD', 'city:houses', n, u);                     // the map reads
this list
  return { house, fresh: true };
}
```



THE REAL PAGES: THE LOCKED NAME, THE HOUSE LANDING ON THE MAP, THE HOUSE PAGE

LIGHT ON PHONES

Most people open it from the DM, on a phone. On phones the city draws half the triangles (fewer segments, no far trees), at 1.5× resolution with smaller shadows, and the generator still runs every random call, so the lots are identical on every device. The page only draws a new frame when something moves.

CHAPTER 08

THE MONEY

UNDER EVERY HOUSE

The house is free. Under it, on the house page, three small offers at a 1,000 followers price, things you can deliver in minutes:

OFFER	WHAT IT IS	PRICE
Make it real	The house as a real photo, one image on Nano Banana Pro from the 3D	€4.99 instead of €49
10-second film	One clip of the house, about 10 credits	€9 instead of €99
Move downtown	A lot by the fountain	€1.99 instead of €19

Above them, for the professionals who came for the reel and not for the house, the free City Blueprint, which leads here. Every Stripe link carries the username, so you know whose house to make.

CHAPTER 09

FOR YOUR BRAND

SAME MACHINE, YOUR CITY

- **A developer:** every follower gets an apartment in the tower; the first ones on the top floors. The reel sells the waiting list.
- **A hotel or a resort:** every follower gets a villa on the island; the early ones on the beach.
- **A town or a neighbourhood:** the real streets from the plan, the real landmarks, the followers as residents.
- **An architect:** your own projects downtown, the followers' houses around them.

CHECK BEFORE YOU POST

- The word in the caption, in the ManyChat trigger and in the reel is the same.
- ManyChat is Live on the next post before you post, not after.
- The secret is set on the site and in the ManyChat header, and one test link opens, reserves and shows the house.
- A second tap on the same link opens the house, it does not make a new one.
- The map opens fast on a phone on 4G, from the DM.

YOUR TURN

NOW BUILD YOUR CITY

Want us to build it for your brand? Write to us on Instagram, **@omnyalab**, or at omnyalab.com.